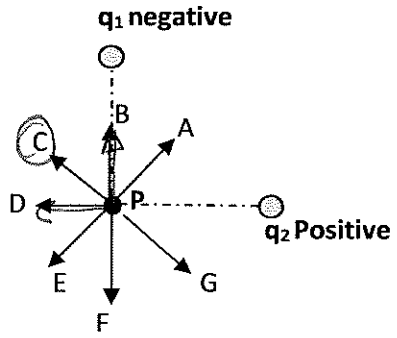
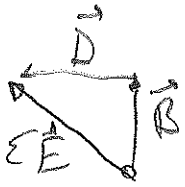
Worksheet # 2 Part A - Multiple Choice
Calculus Physics 1

Name: Key Date: _____

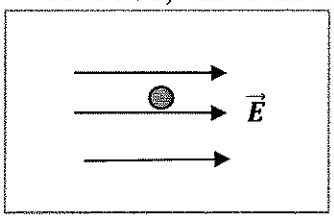
Questions 1 though 10

1) Two spheres with a net charge, each with the same magnitude but varying types are shown (q_1 is negative and q_2 is positive). Which of the arrows labeled **A**, **B**, **C**, and **D** gives the correct direction of the Electric field at point P?





2) A ball has a positive net charge of q = 0.1C and a mass of 2 kg, it is placed in an electric field with a field strength of 60 N/C, what is the acceleration of the ball if no other forces are acting except the electrostatic force?



$\vec{F}_E = q\vec{E}$

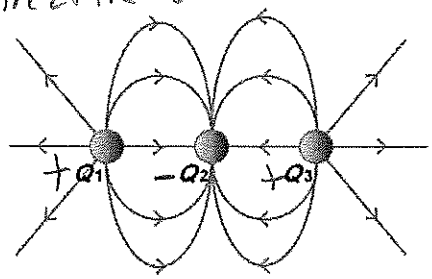
$\Sigma F_x = |\vec{F}_E| = ma_x$

$a_x = \frac{|\vec{F}_E|}{m} = \frac{q|\vec{E}|}{m}$

$a_x = 3\,m/s^2$

3) The figure shows three electric charges labeled Q_1, Q_2, Q_3, and some electric field lines in the region surrounding the charges. What are the signs of the three charges? Direction of Field lines

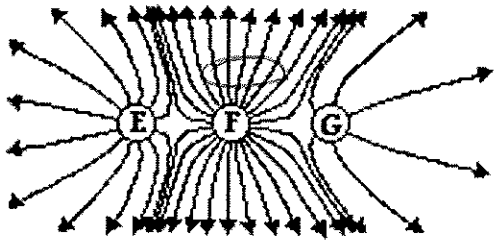
A) Q_1 is positive, Q_2 is negative, Q_3 is positive.
B) Q_1 is negative, Q_2 is positive, Q_3 is negative.
C) Q_1 is positive, Q_2 is positive, Q_3 is negative.
D) All three charges are negative.
E) All three charges are positive.



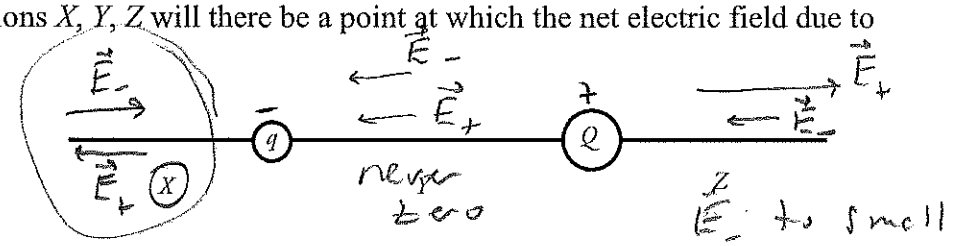
4) Which of the objects shown with a net positive charge has the largest amount of charge?

Larger # of Field lines - Stronger Field

A) E
B) F
C) G
D) They are all the same amount of charge.
E) Can not be determined.



5) The figure shows two unequal point charges, q and Q, of opposite type. Charge Q has greater magnitude than charge q. In which of the regions X, Y, Z will there be a point at which the net electric field due to these two charges is zero?



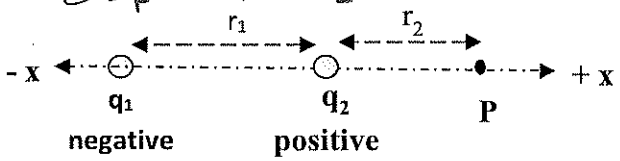
A) only regions X and Z
(B) only region X
C) only region Y
D) only region Z
E) all three regions.
F) No regions.

6) What is the magnitude and direction of the net electric field at point P due to q_1 and q_2 if $q_1 = 20$ μC negative, $q_2 = 20$ μC positive, $r_1 = 1$ m and $r_2 = 0.75$ m?

$|\vec{E}_1| = \frac{kq_1}{(r_1+r_2)^2} = 58{,}775\ N/C$

$|\vec{E}_2| = \frac{kq_2}{r_2^2} = 320{,}000\ N/C$

$\Sigma \vec{E}_p = \vec{E}_1 + \vec{E}_2$



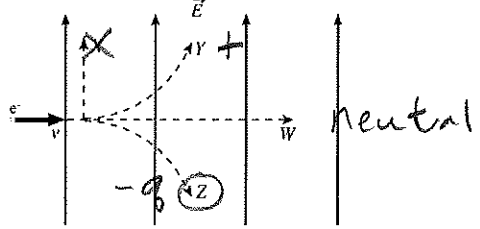
positive x - direction

$\Sigma E_x = (-58775\ N/C) + (320{,}000\ N/C) = +261{,}225\ N/C$

7) An electron is initially moving to the right when it enters a uniform electric field directed upwards. Which trajectory shown below will the electron follow?

A) trajectory W
B) trajectory X
C) trajectory Y
(D) trajectory Z

opposite direction as $\vec{E}$ - field



8) A uniform conducting sphere with a radius of 2 cm has a surface charge density of $\sigma = 4 \times 10^{-6}$ C/m^2, determine the total charge on the sphere.

$\sigma = \frac{Q}{A} = \frac{Q}{4\pi r^2}$ | $Q = \sigma(4\pi r^2) = 2.0 \times 10^{-8} C$

9) Which of the following statements about Gauss's law is correct?

A) Gauss's law is valid only for symmetric charge distributions, such as spheres and cylinders.
B) If there is no charge inside of a Gaussian surface, the electric field must be zero at points of that surface.
C) Only charge enclosed within a Gaussian surface can produce an electric field at points on that surface.
(D) The electric flux passing through a Gaussian surface depends only on the amount of charge inside that surface, not on its size or shape.

10) A sensitive electronic device is placed inside a hollow, uncharged metal conducting box. A strong external electric field is then applied to the box. What happens to the device?

$Q_{in} = 0C$

A) The device is damaged by the induced internal field inside the box.
(B) The device is unaffected because the internal field inside the box remains zero.
C) The device is unaffected because the electric field reflects off the box.
D) The device is only protected if the box is grounded.
E) The external field lines pass through the box but slow down.

$|\vec{E}| = \frac{Q_{in}}{A\varepsilon_0} = 0\ N/C$